

2018-2019 学年第一学期期末考试试卷

一、判断题(每小题 2 分, 共 20 分)

1. () if N numbers are stored in a doubly linked list in increasing order, then the average time complexity for binary search is $O(\log n)$
2. () Mergesort is stable.
3. () If the preorder and the postorder traversal sequences of a binary tree have exactly the opposite orders, then none of the nodes in the tree has two subtrees.
4. () Linear probing is equivalent to double hashing with a secondary hash function of $\text{Hash2}(K) = 1$.
5. () To sort N records, heap sort requires at least $O(N)$ extra space.
6. () $\log N^{20}$ is $O(N)$.
7. () If keys are pushed onto a stack in the order abcde, then it's impossible to obtain the output sequence cedab.
8. () Let P be the shortest path from S to T. If the weight of every edge in the graph is incremented by 1, P will still be the shortest path from S to T.
9. () For a graph, if each vertex has an even degree, we can find an Euler circuit that visits every vertex exactly once.
10. () Given a binary search tree with 20 integer keys which includes 5, 6, and 7, if 5 and 7 are on the same level, then 6 must be their parent.

二、选择题(每小题 3 分, 共 60 分)

1、In order to convert the infix expression $4 + 3 * (6 * 3 - 12)$ to postfix expression using a stack S, then the minimum size of S must be: ()

U. 2

V. 3

W. 4

X. 5

2、Suppose that the size of a hash table is 11, and the hash function is $H(\text{key}) = \text{key} \% 11$. The following 4 elements have been inserted into the table as $\text{Addr}(14)=3$, $\text{Addr}(38)=5$, $\text{Addr}(61)=6$, $\text{Addr}(86)=9$. When open addressing with quadratic probing is used to solve collisions, the address of the element with $\text{key}=49$ will be ()

M. 5

N. 10

O. 7

P. 8

3、 Given input {15, 9, 7, 8 20, -1, 4}. If the result of the 1st run of Shell sort is {15, -1, 4, 8, 20, 9, 7}, then the initial increment must be:()

U. 1

V. 2

W. 3

X. 4

4、 Which one of the following is a possible postorder traversal sequence of a binary search tree?()

U. 3 1 6 2 5 8 10 9 7 4

V. 3 1 2 6 5 8 10 9 7 4

W. 1 3 2 6 5 8 10 9 7 4

X. 1 3 2 6 8 5 10 9 7 4

5、 Given input {46, 79, 56, 38, 40, 84}. After the first partition(with the left most record as the pivot) of quick sort, the resulting sequence is :()

U. {38, 46, 79, 56, 40, 84}

V. {38, 79, 56, 46, 40, 84}

W. {38, 46, 56, 79, 40, 84}

X. {40, 38, 46, 56, 79, 84}

6、 Following is the C-like pseudo code of a function that takes a Queue as an argument.

```
void foo(Queue Q)
```

```
{
```

```
    Queue Q1 – CreateQueue();
```

```
    while( !IsEmpty(Q))
```

```
    {
```

```
        //dequeue am item from Q    and enqueue it into Q1
```

```
        Enqueue(Q1, Dequeue(Q));
```

```
    }
```

```
    while(!IsEmpty(Q1))
```

```
    {
```

```
        // dequeue an item from Q1 and enqueue it into Q
```

```
        Enqueue(Q, Dequeue(Q1));
```

```
    }
```

```
    DisposeQueue(Q1);
```

}

What does the above function do ?()

M. Removes the last item from Q

N. Keeps Q unchanged

O. Makes Q empty

P. Reverses Q

7、 Let T be a tree of N nodes created by union-by-size without path compression, then the minimum depth of T may be()

U. 1

V. $\log N$

W. $N - 1$

X. $N / 2$

8、 For an in-order threaded binary tree, if the pre-order and in-order traversal sequences are D A B C F E and B A C D E F respectively, which pair of nodes' left links are both threads? ()

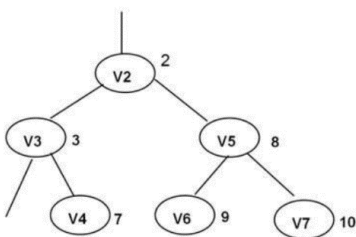
A. D and A

B. A and F

C. C and F

D. C and E

9、 The following is the part of depth-first search tree to find the articulation points, and the Num(v) value has been marked beside each vertex v. The back edges are not shown. Which of the following situation is impossible ()



U. low(v6) is greater than low(v7)

V. low(v4) is 2

W. low(v6) is 3

X. low(v7) is 2

10、 Insert {18, 23, 4, 26, 31, 33, 17, 39} one by one into an initially empty hash table of size 13 with the hash function $H(\text{key}) = \text{key} \% 13$, and linear probing is used to resolve collisions. What is the loading density when the first collision occurs?()

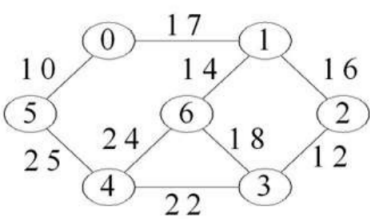
M. 0.54

N. 0.63

O. 0.31

P. 0.62

11、 To find the minimum spanning tree with Prim’s algorithm for the following graph, a sequence of vertices 6, 1, 2, 3 was found during the algorithm’s early steps. Which one vertex will be added in the next step? ()



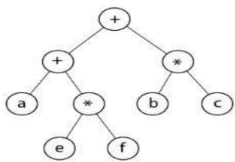
M. 0

N. 4

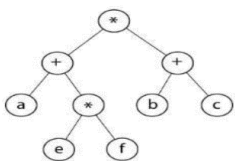
O. 5

P. The vertex serial is incorrect

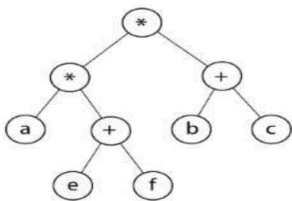
12、 Which one of the following is the expression tree corresponding to the postfix expression aef^*+bc^*+ ()



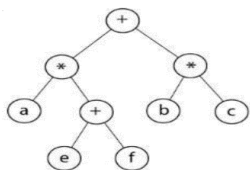
U.



V.



W.



X.

13、It is known that a 3-heap is a heap whose nodes have 3 children. Suppose that the level-order traversal sequence of a max-3-heap is {88, 76, 65, 82, 68, 46, 52, 44, 62, 33, 75, 60, 55, 28}. Use the linear algorithm to adjust this max-3-heap into a min-3-heap, and then run DeleteMin. As a result, there are ___ nodes whose positions are not moved in the process. ()

I. 2

J. 3

K. 4

L. 5

14、A graph with 100 vertices and 12 edges must have at most _____ connected component(s). ()

I. 87

J. 88

K. 94

L. 95

15、When inserting a new key K into a binary search tree T with 512 nodes, the worst-case number of comparisons between K and the keys already in T is in the range of: ()

I. [10, 511]

J. [9, 511]

K. [9, 512]

L. [10, 512]

16、Given a tree of degree 6. Suppose that the numbers of nodes of degrees 1, 2, 3, 4, 5, 6 are 3, 5, 1, 2, 4, 3, respectively. Then the number of leaf nodes must be: ()

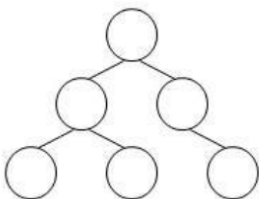
I. 35

J. 39

K. 43

L. 45

17、Given the structure of a binary search tree (as shown in the figure), which one of the following insertion sequences is impossible? ()



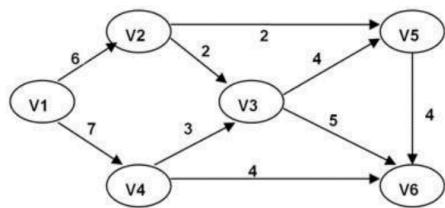
I. 83 67 91 98 20 75

J. 83 67 75 91 10 98

K. 83 91 75 67 20 98

L. 83 91 98 67 75 20

18、 The maximum flow from v1 to v6 is: ()



I. 11

J. 12

K. 13

L. 0

19、 The inorder and the postorder traversal sequences of a binary tree are a b c d e f g and a c b g f e d, respectively. Then the preorder traversal sequences is: ()

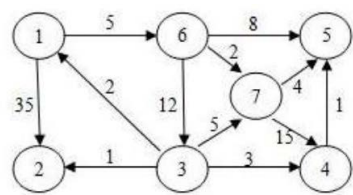
I. d b a c f e g

J. d b a c e f g

K. d a c b f e g

L. d c a b e f g

20、 Use Dijkstra algorithm to find the shortest paths from 1 to every other vertices. In which order that the destinations must be obtained? ()



I. 6, 7, 5, 3, 2, 4

J. 6, 2, 5, 7, 3, 4

K. 2, 3, 4, 5, 6, 7,

L. 2, 4, 3, 6, 5, 7

三、程序填空题（满分 12 分）

1.The function is to find the K-th largest element in a list A of N elements. The function BuildMinHeap(H, K) is to arrange elements H[1] ... H[K] into a min-heap. Please complete the following program.

ElementType FindKthLargest (int A[], int N, int K)

```

{ /* it is assumed that K<=N */
  ElementType *H;
  int i, next, child;
  H = (ElementType *)malloc((K+1)*sizeof(ElementType));
  for ( i=1; i<=K; i++) H[i] = A[i-1];
  BuildMinHeap(H, K);
  for ( next=K; next<N; next++) {
    H[0] = A[next];
    if ( H[0] > H[1] ) {
      for ( i=1; i*2<=K; i=child ) {
        child = i*2;
        if ( child!=K && _____(3 分) ) child++;
        if ( _____(3 分) )
          H[i] = H[child];
        else break;
      }
      H[i] = H[0];
    }
  }
  return H[1];
}

```

2. Given an array $a[]$ of n integers, the function MissingMin is to find and return the minimum possible integer which is NOT in the array. For example, give $\{3, -1, 8, 1, 0\}$, 2 is the smallest positive integer which is missing.

```

int MissingMin(int a[], int n)
{
  int i, j, min, missing = 1;
  for(i = 0; i < n ; i++)
  {
    min = i;
    for(j = i + 1 ; j < n ; j++)
      If(_____ (3 分))min = j;
  }
}

```

```
        if(min != i) swap(a[i], a[min]);

        if(a[i] == missing)missing++;

        else if(a[i] > missing)_____ (3 分);

    }

    return missing;

}
```

四、函数题

Format of functions:

```
bool IsTopSeq(LGraph Graph, Vertex Seq[]);
```

where LGraph is defined as the following :

```
struct AdjVNode{
    Vertex AdjV;
    PtrToAdjVNode Next;
};

typedef struct Vnode{
    PtrToAdjVNode FirstEdge;
}AdjList[MaxVertexNum];
```

```
typedef struct GNode *PtrToGNode;
struct GNode{
    int Nv;
    int Ne;
    AdjList G;
};
```

```
typedef PtrToGNode LGgraph;
```

The function IsTopSeq must return true if Seq does correspond to a topological order; otherwise return false.

Note: Although the vertices are numbered form 1 to MaxVertexNum, they are indexed from 0 in the LGraph structure.

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
typedef enum{false, true} bool;
```

```
#define MaxVertexNum 10 // maximum number of vertices
```

```
typedef int Vertex;//vertices are numberd from 1 to MaxVertexNum
```

```
typedef struct AdjVNode *PtrToAdjVNode;
```

```
struct AdjVNode{
    Vertex AdjV;
    PtrToAdjVNode Next;
};
```

```
typedef struct Vnode{
    PtrToAdjVNode FirstEdge;
}AdjList[MaxVertexNum];
```

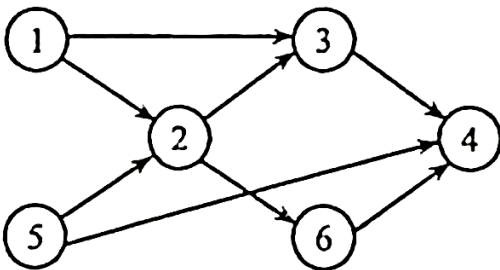
```

typedef struct GNode *PtrToGNode;
struct GNode{
    int Nv;
    int Ne;
    AdjList G;
};
typedef PtrToGNode LGraph;
LGraph ReadG();//details omitted
bool IsTopSeq(LGraph Graph, Vertex Seq[]);
int main()
{
    int i, j, N;
    Vertex Seq[MaxVertexNum];
    LGraph G = ReadG();
    scanf("%d", &N);
    for(i = 0; i < N ; i ++)
    {
        for(j = 0 ; j < G->Nv ; j ++)
            scanf("%d", &Seq[j]);
        if(IsTopSeq(G, Seq) == true)print("yes\n");
        else printf("no\n");
    }
    return 0;
}

```

//Your function will be put here

Sample Input(for the graph shown in the figure):



```

6 8
1 2
1 3
5 2
5 4
2 3
2 6
3 4
6 4
5
1 5 2 3 6 4
5 1 2 6 3 4

```

《数据结构基础》历年题

5 1 2 3 6 4

5 2 1 6 3 4

1 2 3 4 5 6

Sample Output:

yes

yes

yes

no

no

2018-2019 学年第一学期期末考试试卷参考答案

一、判断题(每小题 2 分, 共 20 分)

1、【正解】F

【解析】双向链表不支持随机访问, 因此在二分查找时, 中点元素需要线性的时间复杂度来访问, 所以平均时间复杂度为 $O(n)$

【考点延伸】双向链表

2、【正解】T

【解析】归并排序是稳定的排序算法

【考点延伸】归并排序

3、【正解】T

【解析】前序和后序序列如果是相反的, 则每一个结点一定只有 1 棵子树或没有子树

【考点延伸】二叉树的遍历

4、【正解】T

【解析】线性探测本质上可以看作一个多次哈希的过程, 因为每次发生冲突后, 都是往后移动一位, 看在此位置时是否发生冲突。

【考点延伸】散列冲突

5、【正解】F

【解析】堆排序在调整的过程中不需要额外的空间, 因此空间复杂度为 $O(1)$

【考点延伸】堆排序

6、【正解】F

【解析】根据对数的运算法则, $\log N^{20} = 20 \log N$, 因此时间复杂度为 $O(\log N)$

【考点延伸】时间复杂度

7、【正解】T

【解析】在 d 出栈后, 栈中的序列一定为 ab, 因此后序的出栈序列一定为 ba, 不可能为 ab

【考点延伸】栈

8、【正解】F

【解析】举个反例, 假设从 S 到 T 共有两条路径, 7 和 2, 2, 2, 此时最短路径为 2, 2, 2; 当每一条边都增加 1 时两条路径变为 8 和 3, 3, 3, 此时最短路径为 8

【考点延伸】最短路径

9、【正解】F

【解析】无向图存在欧拉回路的充要条件: 一个无向图存在欧拉回路, 当且仅当该图所有顶点度数都为偶数, 且该图是连通图。有向图存在欧拉回路的充要条件: 一个有向图存在欧拉回路, 所有顶点的入度等于出度且该图是连通图。题中给的图不明确, 因此无法判断是否具有欧拉回路

【考点延伸】欧拉回路

10、【正解】F

【解析】5, 7 即使在同一层, 也不一定是同一个父结点的孩子

【考点延伸】二叉搜索树

二、单选题(每小题 3 分, 共 60 分)

1 【正解】C

【解析】转化过程: 初始化两个栈: 运算符栈 S1 和储存中间结果的栈 S2;

从左至右扫描中缀表达式;

遇到操作数时, 将其压入 S2;

遇到运算符时, 比较其与 S1 栈顶运算符的优先级;

如果 S1 为空, 或栈顶运算符为左括号“(”, 则直接将此运算符入栈;

否则, 若优先级比栈顶运算符的高, 也将运算符压入 S1 (注意转换为前缀表达式时是优先级较高或相同, 而这里则不包括相同的情况);

否则, 将 S1 栈顶的运算符弹出并压入到 S2 中, 再次转到(4-1)与 S1 中新的栈顶运算符相比较;

遇到括号时:

如果是左括号“(”, 则直接压入 S1;

如果是右括号“)”, 则依次弹出 S1 栈顶的运算符, 并压入 S2, 直到遇到左括号为止, 此时将这一对括号丢弃;

循环扫描, 直到表达式的最右边;

将 S1 中剩余的运算符依次弹出并压入 S2;

在转化的过程中栈 S1 所占用的最大空间为 4

【考点延伸】后缀表达式的转化

2 【正解】B

【解析】 $49 \% 11 = 5$, 发生冲突

第一次探测, $5+1=6$, 冲突

第二次探测, $6+4=10$, 退出探测

【考点延伸】平方探测

3 【正解】D

【解析】可以观察得出, 15, 20 和 9, -1 和 7, 4 的进行了排序, 因此增量为 4

【考点延伸】希尔排序

4 【正解】C

【解析】二叉搜索树满足任意一个结点的值大于其左孩子的值而小于其右孩子的值(如果左右孩子存在的话)根据后序遍历序列和二叉搜索树的特性可以构造出完整的二叉树, 只有 C 是符合条件的

【考点延伸】二叉搜索树

5 【正解】D

【解析】所选取的 pivot 为 40, 第一趟排序后的结果为 40, 38, 46, 56, 79, 84

【考点延伸】快速排序

6 【正解】B

【解析】该函数先让队列 Q 中的元素出队后入队 Q1 中, 然后又让 Q1 中的元素出队后入队 Q, Q 中的元素没有发生任何变化

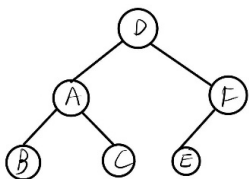
【考点延伸】队列

7 【正解】A

【解析】最小深度对应的情况是: N-1 个结点直接连接在根结点上, 此时深度为 1

【考点延伸】树

8 【正解】D



【解析】

【考点延伸】希尔排序

9 【正解】B

【考点延伸】深度优先搜索

10 【正解】C

【解析】在第一次冲突时，装载率为 $4/13=0.31$

【考点延伸】散列冲突

11 【正解】A

【解析】和 6, 1, 2, 3 相连且不会出现闭环的边权重最小的是 17, 0-1, 所以下一个被加入的顶点为 0

【考点延伸】Prim 算法

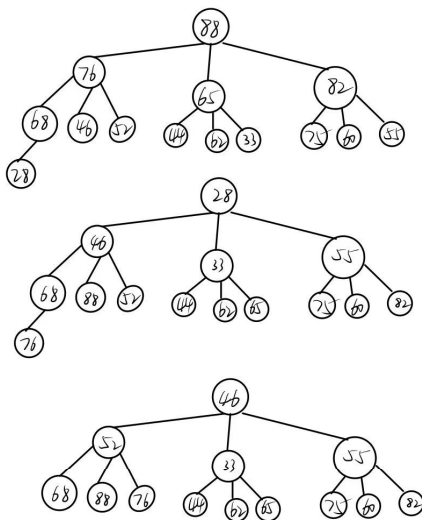
12 【正解】A

【解析】后缀表达式就是表达式树的后序遍历序列

【考点延伸】后缀表达式

13 【正解】C

共有 4 个元素位置未发生变化



【解析】

【考点延伸】堆的调整

14 【正解】D

【解析】当 6 个顶点用 12 条边连接时，剩下 94 个顶点组成 94 个连通图，外加 1 个共 95 个连通图

【考点延伸】无向图

15 【正解】D

【解析】512 个结点最大深度为 512，最小深度为 10，因此最坏情况下需要比较的区间范围为 [10, 512]

【考点延伸】二叉搜索树

16 【正解】D

【解析】由等式顶点数=边数+1 可得，叶子结点数为 45

【考点延伸】树的度

17 【正解】C

【解析】C 选项在插入 2 时与给出的结构不同

【考点延伸】二叉搜索树

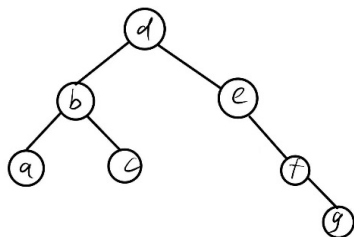
18 【正解】A

【解析】从 v_1 同时出发的流量 6 和 7 最多只能有 11 到达 v_6

【考点延伸】最大流

19 【正解】B

【解析】由中序遍历和后序遍历可得二叉树如下：



【考点延伸】二叉树的遍历

20 【正解】A

【解析】每次都把 dist 数组中值最小的顶点加入已收集结点集合，然后判断 1 通过该顶点到其它顶点是否缩短了距离，如果缩短就改变 dist 数组的值。因此被加入已收集集合的顶点顺序是 6 7 5 3 2 4

【考点延伸】Dijkstra 算法

三、程序填空题（满分 12 分）

1、【解析】(1) $H[\text{child}] > H[\text{child} + 1]$

(2) $H[\text{child}] < H[0]$

【考点延伸】堆

2、【解析】(1) $a[j] < a[\text{min}]$

(2) break

【考点延伸】简单选择排序变形

四、函数题(满分 8 分)

【解析】

```
bool IsTopSeq(LGraph Graph, Vertex Seq[]){
    int degree[MaxVertexNum];
    for(int i; i < Graph->Nv; i++){
        degree[i]=0;
    }
    int i;
    PtrToAdjVNode p;
    for(i = 0; i < Graph->Nv; i++){
        for(p=Graph->G[i].FirstEdge; p=p->Next)
        {
            degree[p->AdjV]++;
        }
    }
}
```

```

int j;
for(i = 0; i < Graph->Nv;i++){
    if(degree[Seq[i]-1]!=0)
        return 0;
    else{
        for(p=Graph->G[Seq[i]-1].FirstEdge;p=p->Next)
            degree[p->AdjV]--;
    }
}
return 1;
}

```

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